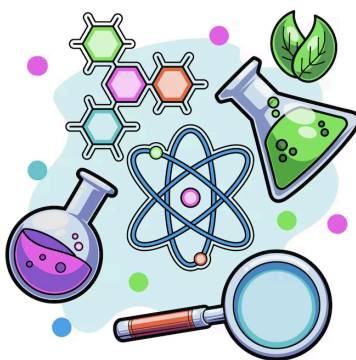




UNIVERSITY INTERSCHOLASTIC LEAGUE

Science

Invitational B • 2026



GENERAL DIRECTIONS:

- **DO NOT OPEN EXAM UNTIL TOLD TO DO SO.**
- Contestants may take up to two hours to complete the contest. If you are in the process of actually writing an answer when the signal to stop is given, you may finish writing that answer.
- Papers may not be turned in until 30 minutes have elapsed. If you finish the test in less than 30 minutes, remain at your seat and retain your paper until told to do otherwise. You may use this time to check your answers.
- All answers must be written on the answer sheet provided. Indicate your answers in the appropriate blanks provided on the answer sheet. Write clearly and legibly!
- You may place as many notations as you desire anywhere on the test paper but not on the answer sheet, which is reserved for answers only.
- You may use additional scratch paper provided by the contest director.
- All questions have ONE and only ONE correct (BEST) answer. There is a penalty for all incorrect answers.
- If a question is omitted, no points are given or subtracted.
- The back two pages of this test include a copy of the periodic table of the elements, as well as listings of other scientific relationships. You may use this information during the contest and may detach the back page from the test if you wish.
- A simple scientific calculator is sufficient for the high school Science contest. **The UIL provides a list of approved calculators that meet the criteria for use in the Science contest. No other calculators are permitted during the contest.** The Science Contest Approved Calculator List is available in the current Science Contest Handbook and on the UIL website. Contest directors will perform a brief visual inspection to confirm that all contestants are using only approved calculators. Each contestant may use up to two approved calculators during the contest.

- B01. In incomplete dominance, what is observed in the heterozygous phenotype?
- A) Only the dominant trait appears.
 - B) Both alleles are expressed fully and separately.
 - C) The phenotype is a blend of the two alleles.
 - D) The recessive allele is silenced by methylation.
 - E) The heterozygous individual is sterile.

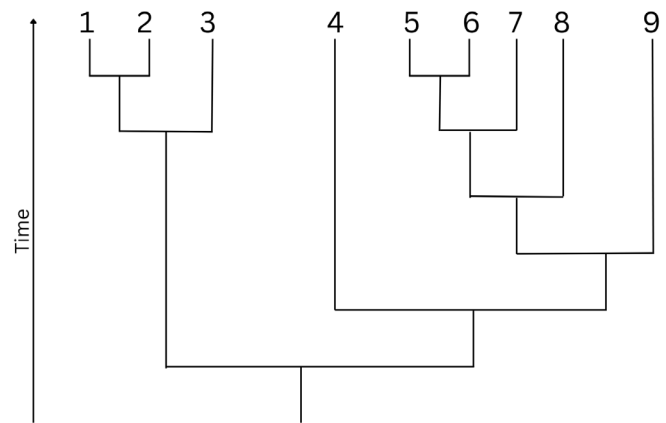
- B02. Which of the following is an ecological role of bacteria?
- A) Primary producers
 - B) Pathogens
 - C) Decomposers
 - D) Nitrogen fixation
 - E) All the above

- B03. Which of the following regulates progression through cell cycle checkpoints such as G1 and G2?
- A) Histone acetylation and DNA methylation.
 - B) Cyclins and cyclin-dependent kinases (CDKs).
 - C) Spindle fiber length and centrosome duplication.
 - D) Telomerase activity and ribosome number.
 - E) Mitochondrial membrane potential and oxygen availability.

- B04. In an experiment, a group of cells placed in a hypertonic solution shrivel and shrink. Which cellular structure is most responsible for controlling water movement into and out of the cell?
- A) Plasma membrane
 - B) Mitochondrial membrane
 - C) Nuclear envelope
 - D) Endoplasmic reticulum
 - E) Cell wall

- B05. Which of the following best describes a homologous structure?
- A) Traits that serve the same function but have different evolutionary origins.
 - B) Traits that appear identical in all species.
 - C) Anatomical features that are inherited from a common ancestor.
 - D) Traits that arise due to similar environmental pressures in unrelated species.
 - E) Fossils with no clear modern counterparts.

- B06. Examine the phylogenetic tree below. Which species shares a more recent common ancestor with species 7?



- A) 1
- B) 3
- C) 4
- D) 8
- E) 9

- B07. What is the primary mode of transmission for *Bordetella pertussis*?
- A) Respiratory droplets
 - B) Contaminated food and water
 - C) Airborne spores
 - D) Bloodborne exposure
 - E) Contact with contaminated surfaces

- B08. In guinea pigs, black fur (B) is dominant over white fur (b). Two heterozygous black guinea pigs are crossed. Out of 500 offspring, approximately how many would you expect to have white fur?
- A) 0
 - B) 125
 - C) 250
 - D) 375
 - E) 500
- B09. Which of the following plant responses is regulated by phototropism?
- A) The closing of stomata during a drought.
 - B) The movement of chloroplasts toward the cell wall.
 - C) The growth of a plant stem toward light.
 - D) The downward growth of roots in soil.
 - E) The opening of flowers at night.
- B10. Which organ system protects the body from external pathogens and helps regulate body temperature?
- A) Respiratory
 - B) Circulatory
 - C) Digestive
 - D) Nervous
 - E) Integumentary
- B11. A researcher discovers a molecule made of carbon, hydrogen, and oxygen in a 1:2:1 ratio. It is readily soluble in water and breaks down to release energy. This molecule is most likely a
- A) monosaccharide.
 - B) steroid.
 - C) polypeptide.
 - D) nucleic acid.
 - E) phospholipid.
- B12. In November 2025, the Center for Disease Control and Prevention and public health officials investigated an ongoing multistate outbreak of _____ linked to infant formula.
- A) *Salmonella*
 - B) Hepatitis B
 - C) *Listeria*
 - D) *E. coli*
 - E) botulism
- B13. During cellular respiration in eukaryotes, the electron transport chain and ATP synthase are embedded within which structure, allowing chemiosmosis to occur across it?
- A) Chloroplast outer membrane
 - B) Thylakoid membrane
 - C) Mitochondrial matrix
 - D) Inner mitochondrial membrane
 - E) Nuclear envelope
- B14. A competitive inhibitor binds to the site on an enzyme where the substrate normally attaches. How would this affect the enzyme's function?
- A) It blocks substrate binding by occupying the active site.
 - B) It changes the enzyme's shape, preventing substrate binding at a different site.
 - C) It increases enzyme activity by stabilizing the substrate.
 - D) It permanently deactivates the enzyme by destroying the active site.
 - E) It binds the substrate, enhancing the reaction rate.
- B15. Which of the following best describes an operon that is normally inactive, but can be turned on in the presence of a specific substrate?
- A) Repressible operon
 - B) Inducible operon
 - C) Promoter
 - D) Operator
 - E) Constitutive operon

- B16. A small group of birds is blown off course onto a new island, where they establish a population. Over time, this population shows reduced genetic variation compared to the mainland. This is best explained by:
- A) Stabilizing selection
 - B) Founder effect
 - C) Mutation
 - D) Gene flow
 - E) Overdominance
- B17. What does population density refer to?
- A) The number of individuals per unit area.
 - B) The total number of individuals in a population.
 - C) The genetic diversity within a population.
 - D) The rate at which a population grows over time.
 - E) The number of species in a community.
- B18. Negative feedback systems are the primary way that the body maintains stability, known as ____.
- A) metabolism
 - B) growth
 - C) reproduction
 - D) differentiation
 - E) homeostasis
- B19. Which of the following best describes the lytic cycle of a virus?
- A) The virus integrates its genome into the host's DNA and remains dormant.
 - B) The virus hijacks host machinery to produce new viral particles and bursts the host cell.
 - C) The virus replicates inside the host without causing cell death.
 - D) The virus causes the host cell to divide uncontrollably, forming a tumor.
 - E) The virus triggers the host immune system to destroy infected cells before replication.
- B20. What happens immediately after the ribosome encounters a stop codon during translation?
- A) A tRNA recognizes the stop codon and adds the stop amino acid.
 - B) The ribosome recruits RNA polymerase to restart translation.
 - C) The ribosome detaches and releases the completed polypeptide.
 - D) The mRNA is degraded and recycled.
 - E) The ribosome begins transcribing the next gene.

C01. How many atoms are there in one formula unit of potassium carbonate?

- A) 3 B) 4 C) 5 D) 6 E) 7

C02. How many oxygen molecules are needed when the equation for the formation of iron(III) oxide from elemental iron and oxygen is balanced using the smallest whole numbers?

- A) 2 B) 3 C) 4 D) 5 E) 6

C03. What is the difference, if any, between atomic mass and mass number?

- A) Atomic mass and mass number are different terms for the same thing.
 B) Mass number is an obsolete term for atomic mass and is no longer used.
 C) Atomic mass is the decimal value at the bottom of a periodic table entry. Mass number is the integer at the top of the periodic table entry.
 D) Atomic mass is specifically the mass of an atom while mass number can be the mass of either an atom or a molecule.
 E) Atomic mass is the average mass of all isotopes of an element while mass number is the number of nucleons in one specific isotope of that element.

C04. How many of these molecules contain π bonds?

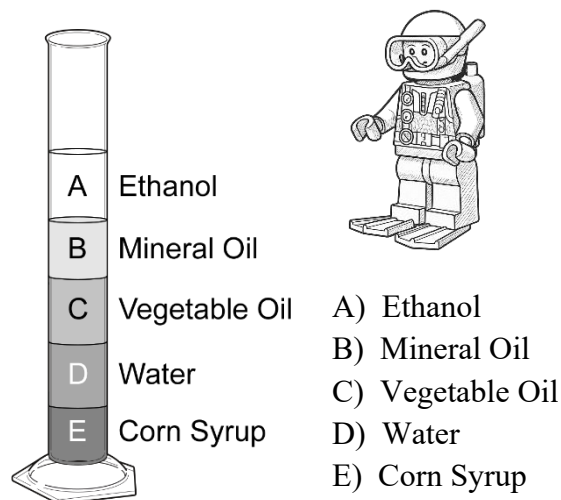
CO₂ CH₄ H₂O NH₃ CO

- A) 1 B) 2 C) 3 D) 4 E) 5

C05. If 22.4 grams of dry ice (frozen carbon dioxide) is allowed to sublime in a flexible-walled container to 25°C at 1.0 atm pressure, what will the volume of the gas be?

- A) 1.0 L
 B) 6.2 L
 C) 12.4 L
 D) 18.6 L
 E) 22.4 L

C06. A graduated cylinder is filled with samples of five different liquids. A Lego SCUBA diver is dropped into the cylinder and begins to sink. Which is the lowest layer that the diver will sink to the bottom of?



A) Ethanol
 B) Mineral Oil
 C) Vegetable Oil
 D) Water
 E) Corn Syrup

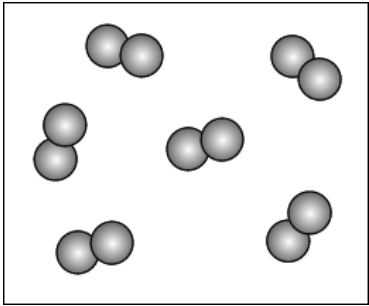
C07. The ΔH for the reaction between HCl and sodium bicarbonate is 28.5 kJ/mol. This reaction is best described as

- A) endothermic
 B) exothermic
 C) ectothermic
 D) isothermic
 E) xenomorphic

C08. In which of these chemical reactions is the carbon atom being oxidized?

- (i) $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$
 (ii) $\text{CH}_4 + 4\text{Cl}_2 \rightarrow \text{CCl}_4 + 4\text{HCl}$
 (iii) $2\text{CO} + \text{O}_2 \rightarrow 2\text{CO}_2$
 (iv) $\text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{CO}_3$

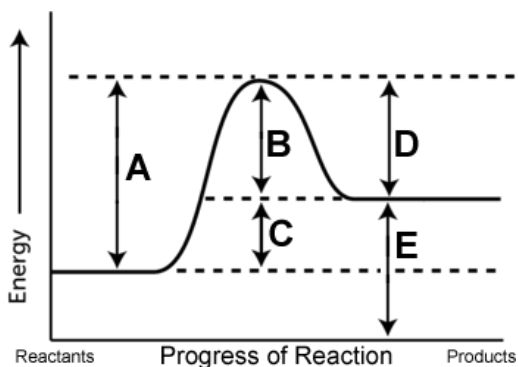
- A) i and iii
 B) ii and iv
 C) i, ii, and iii
 D) i, ii, and iv
 E) all of them

- C09. Which statement below about phase changes of water is true?
- A) Melting 10 g of ice requires as much heat as is given off when 10 g of water freezes.
 - B) It takes more heat to melt 10 g of ice to water than it takes to boil 10 g of water to vapor.
 - C) It takes the same amount of heat to melt 10 g of ice to water as it does to boil 10 g of water to vapor.
 - D) Freezing liquid water to ice absorbs heat from the surroundings, which is why ice makes things cold.
 - E) Boiling requires more energy than freezing because the H_2O molecules have to be broken into H_2 and O_2 .
- C10. Which of the changes below will change the value for the equilibrium constant K for a reaction?
- A) Increasing the pressure
 - B) Raising the temperature
 - C) Adding more reactant
 - D) Adding more product
 - E) All of these
- C11. While doing your chemistry homework you need to know the dissociation constant (K_a) for hypochlorous acid, but the Wikipedia article for hypochlorous acid doesn't include the K_a . Instead it lists the $\text{p}K_a$ as 7.53. How would you determine the K_a for this acid from $\text{p}K_a$?
- A) $K_a = \text{p}K_a$
 - B) $K_a = -\log(\text{p}K_a)$
 - C) $K_a = -\ln(\text{p}K_a)$
 - D) $K_a = e^{\text{p}K_a}$
 - E) $K_a = 10^{-\text{p}K_a}$
- C12. Which of these situations is an example of the common ion effect affecting the solubility of the added salt?
- A) Adding solid NaCl to a 0.10 M solution of NaCl
 - B) Adding solid AgCl to a saturated solution of AgCl
 - C) Adding solid AgCl to a 0.10 M solution of NaCl
 - D) Adding solid NaCl to a saturated solution of AgCl
 - E) All of these
- C13. If you want to determine how much of a reactant remains after a chemical reaction is allowed to run for a period of time, which of these would you use to calculate it?
- A) the equilibrium expression for the reaction
 - B) Kinderhook's Law
 - C) the rate law for the reaction
 - D) the integrated rate law for the reaction
 - E) the Arrhenius equation
- C14. Which term below best describes the gas sample shown here?
- 
- A) Molecules
 - B) A compound
 - C) A mixture of atoms
 - D) A mixture of compounds
 - E) A mixture of elements

C15. What is the pH of a 175 mL solution of 0.75 M NH_3 ? K_b for $\text{NH}_3 = 4.75 \times 10^{-5}$.

- A) 11.78
- B) 2.59
- C) 13.12
- D) 12.33
- E) 12.65

C16. In the potential energy diagram shown below, which arrow or sum of arrows tells you whether the reaction is endothermic or exothermic?



- A) A
- B) B
- C) C
- D) B+C
- E) B+E

C17. When solid calcium carbonate is heated, the products are carbon dioxide and calcium oxide. What type of reaction is this?

- A) Decomposition
- B) Synthesis
- C) Combustion
- D) Double Replacement
- E) Single Replacement

C18. What is the percent copper by mass in CuCl_2 ?

- A) 33.3
- B) 52.7
- C) 35.8
- D) 64.2
- E) 47.3

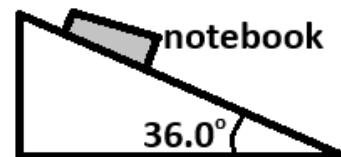
C19. What is the K_{sp} expression for the precipitation reaction between aqueous $\text{Ba}(\text{NO}_3)_2$ and aqueous NaBrO_3 ?

- A) $K_{sp} = \frac{[\text{Na}^+][\text{BrO}_3^-]}{[\text{Ba}^{2+}][\text{NO}_3^-]^2}$
- B) $K_{sp} = \frac{[\text{Ba}^{2+}][\text{NO}_3^-]}{[\text{Na}^+][\text{BrO}_3^-]}$
- C) $K_{sp} = [\text{Ba}^{2+}][\text{BrO}_3^-]^2$
- D) $K_{sp} = \frac{1}{[\text{Ba}^{2+}][\text{BrO}_3^-]^2}$
- E) $K_{sp} = \frac{[\text{Ba}^{2+}][\text{BrO}_3^-]^2}{[\text{Ba}(\text{BrO}_3)_2]}$

C20. One ampere (A) is equal to one coulomb of charge passing a given point each second. If a 2 A electrical current is passed through a solution of Ag^+ for 40 minutes, how many moles of solid Ag would be formed?

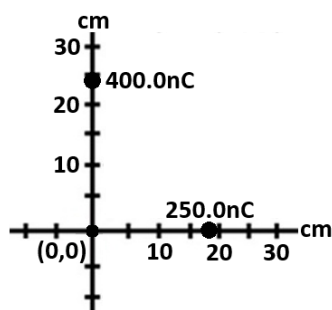
- A) 0.040 mol
- B) 0.050 mol
- C) 0.060 mol
- D) 0.070 mol
- E) 0.080 mol

- P01. According to Malley, Frederick Soddy and William Ramsay isolated the emanations from radium. In the emanation mixture, they observed the spectral lines of which element?
- Actinium
 - Polonium
 - Radon
 - Argon
 - Helium
- P02. According to Malley, Paul Villard sent radium rays through a magnetic field, which broke the rays into different parts. Which type of radiation did Villard discover by analyzing the different parts of the radium rays?
- Alpha particles
 - Beta particles
 - Gamma rays
 - Delta rays
 - Neutrinos
- P03. According to Malley, Norman Campbell and Albert Wood claimed to detect weak radioactivity from elements that are very different from uranium, radium, and thorium. Which of these is an element about which they made this claim?
- Helium
 - Lead
 - Carbon
 - Potassium
 - Germanium
- P04. Which class of main sequence stars spends the longest amount of time on the main sequence?
- Class O blue stars
 - Class A white stars
 - Class G yellow stars
 - Class K orange stars
 - Class M red stars
- P05. A leaf blowing in the wind floats horizontally across a football field (goal post to goal post, 120 yards.) The leaf takes 12.5 minutes to travel the entire distance. What is the speed of the leaf in miles per hour?
- 0.109 mph
 - 0.160 mph
 - 0.224 mph
 - 0.327 mph
 - 0.480 mph
- P06. While trying to crack a walnut, you climb up a ladder and throw the walnut straight down onto a concrete floor. You throw the walnut from a height of 2.35m, and the walnut hits the ground exactly 0.250 seconds after you throw it. With what initial velocity did you throw the walnut?
- 9.40 m/s
 - 8.18 m/s
 - 7.67 m/s
 - 5.92 m/s
 - 4.17 m/s
- P07. A 1.50kg notebook is placed on a frictionless inclined plane. The plane is angled at 36.0° with respect to horizontal, and the notebook slides down the incline as expected. While it is on the incline, what is the magnitude of the normal force (F_N) acting on the notebook?
- 14.7 N
 - 11.9 N
 - 8.64 N
 - 7.12 N
 - 5.76 N



- P08. A 600g toy cart starts at rest on a frictionless horizontal track. You throw a 350g blob of clay, which collides with the back of the cart and sticks to it. Just before the collision, the clay has a horizontal velocity of 13.0m/s. After the collision, the cart-clay system slides forward along the track. What is the speed of the cart-clay system after the collision?
- 4.79 m/s
 - 5.72 m/s
 - 6.65 m/s
 - 7.58 m/s
 - 8.21 m/s
- P09. A merry-go-round starting from rest slowly accelerates to an angular velocity of 9.00rad/s . The merry-go-round takes two complete revolutions to reach its maximum angular velocity. What is the average angular acceleration of the merry-go-round?
- 0.716 rad/s^2
 - 2.05 rad/s^2
 - 3.22 rad/s^2
 - 6.44 rad/s^2
 - 20.3 rad/s^2

- P10. Wind blows across a hollow log that is open on both ends, producing an eerie sound at a frequency of 144 Hz. You measure the log to be 4.76 meters long. Which harmonic is resonating in the log? Assume the air temperature is 20.0°C.
- the $n = 1$ fundamental harmonic
 - the $n = 2$ harmonic
 - the $n = 3$ harmonic
 - the $n = 4$ harmonic
 - the $n = 5$ harmonic
- P11. A cube of aluminum has a mass of 2.80kg. What is the apparent weight of the cube when it is completely submerged in pure water? The density of aluminum is 2700kg/m³.
- 10.2 N
 - 17.3 N
 - 22.4 N
 - 27.4 N
 - 37.6 N
- P12. You explore the electrical properties of dry pasta by connecting one strand of spaghetti to a 24.0V power supply. The current flowing through the spaghetti is measured to be 56.0nA. The cylindrical strand of spaghetti is 36.0cm long and has a diameter of 1.20mm. What is the resistivity of the strand of dry spaghetti?
- 1350 Ωm
 - 2930 Ωm
 - 4300 Ωm
 - 5380 Ωm
 - 7330 Ωm
- P13. Two charges are placed as illustrated. A +400.0nC charge is placed at (0.0, 24.0cm) and a +250.0nC charge is placed at (18.0cm, 0.0). What is the electric potential (voltage) at the origin (0.0, 0.0) due to these two charges?
- 2.50 kV
 - 6.94 kV
 - 9.99 kV
 - 27.5 kV
 - 132 kV
- P14. A particle with the same charge as a proton is moving at a speed of 4.50×10^6 m/s. The particle enters a region where there is a 0.780T magnetic field that is oriented perpendicular to the motion of the particle. The circular path traced out by the particle in the field has a radius of 2.20m. What is the mass of the particle as compared to the mass of a proton?
- $6.11m_p$
 - $18.3m_p$
 - $36.5m_p$
 - $51.8m_p$
 - $67.1m_p$
- P15. An unpolarized beam of light with an initial intensity of 600W/m² passes through a polarizer whose axis is oriented horizontally. After passing through that first polarizer, the beam goes through a second polarizer whose axis is oriented at 30.0° with respect to horizontal. What is the intensity of the beam after passing through both polarizers?
- 75.0 W/m²
 - 150 W/m²
 - 225 W/m²
 - 260 W/m²
 - 450 W/m²
- P16. You look into the still water of a quiet pond and see a minnow swimming just below the surface. The minnow *appears* to be swimming at a point that is 12.0cm deep. How deep below the surface is the minnow *actually* swimming? The index of refraction of water is 1.33.
- 8.00 cm
 - 9.00 cm
 - 12.0 cm
 - 13.3 cm
 - 16.0 cm



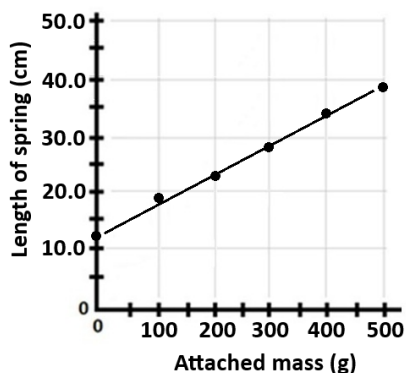
P17. While studying the spectrum of hydrogen, you observe a spectral line with a wavelength of 1005nm. Which transition in hydrogen produces this spectral line?

- A) $n = 6$ to $n = 3$
- B) $n = 7$ to $n = 3$
- C) $n = 7$ to $n = 4$
- D) $n = 8$ to $n = 3$
- E) $n = 8$ to $n = 4$

P18. The K^+ meson has a strangeness of +1, a charge number of +1, and a spin of 0. Its lifetime is $1.2 \times 10^{-8}s$, and it commonly decays into a pair of pions: $K^+ \rightarrow \pi^+ + \pi^0$. What fundamental force is responsible for the decay of the K^+ meson?

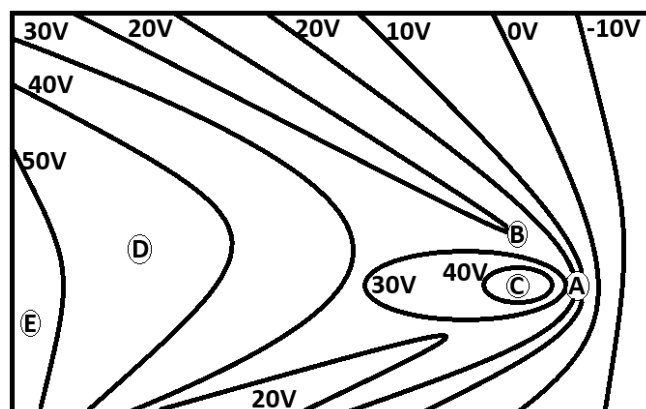
- A) the Weak Force
- B) the Strong Force
- C) the Electromagnetic Force
- D) the Gravitational Force
- E) the Higgs Force

P19. A spring dangles vertically from a hook attached to the ceiling. The unstretched length of the spring is 12.0cm. Various masses are attached to the lower end of the spring, and the length of the spring is measured for each mass. The data are plotted below. Based on these data, what is the approximate spring constant of the spring?



- A) 0.50 N/m
- B) 5.0 N/m
- C) 10 N/m
- D) 13 N/m
- E) 18 N/m

P20. The following map illustrates the equipotential surfaces in a region of space. Based on this map, at what location is the electric field at its greatest magnitude?



- A) location A
- B) location B
- C) location C
- D) location D
- E) location E

Chemistry																8A 18						
1A 1																2 8A 18						
1 H 1.01	2A 2																3A 13	4A 14	5A 15	6A 16	7A 17	He 4.00
3 Li 6.94	4 Be 9.01																5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31	3B 3		4B 4	5B 5	6B 6	7B 7	8B 8 9 10			1B 11	2B 12	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95				
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.64	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80					
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.90	54 Xe 131.29					
55 Cs 132.91	56 Ba 137.33	57 La 138.9	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.20	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)					
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (266)	107 Bh (264)	108 Hs (277)	109 Mt (268)	110 Ds (281)	111 Rg (281)	112 Cn (285)	113 Nh (286)	114 Fl (289)	115 Mc (289)	116 Lv (293)	117 Ts (293)	118 Og (294)					

58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)

Water Data

T_{mp}	= 0°C
T_{bp}	= 100°C
c_{ice}	= 2.09 J/g·K
c_{water}	= 4.184 J/g·K
c_{steam}	= 2.03 J/g·K
ΔH_{fus}	= 334 J/g
ΔH_{vap}	= 2260 J/g
K_f	= 1.86 °C/m
K_b	= 0.512 °C/m

Constants

R	= 0.08206 L·atm/mol·K
R	= 8.314 J/mol·K
R	= 62.36 L·torr/mol·K
e	= 1.602×10^{-19} C
N_A	= 6.022×10^{23} mol ⁻¹
k	= 1.38×10^{-23} J/K
h	= 6.626×10^{-34} J·s
c	= 3.00×10^8 m/s
R_H	= 2.178×10^{-18} J
m_e	= 9.11×10^{-31} kg
$\mathcal{F}$	= 96,485 C/mol e ⁻
1 amp	= 1 C/sec
1 mol e ⁻	= 96,485 C

Densities (g/mL)

Ethanol	0.79
Mineral oil	0.81 – 0.89
Vegetable oil	0.92
Water	1.00
Corn syrup	1.38 – 1.48
Lego piece	1.03 – 1.06

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Physics

Useful Constants

quantity	symbol	value
Free-fall acceleration	g	9.80 m/s^2
Permittivity of Free Space	ϵ_0	$8.854 \times 10^{-12} \text{ C}^2/\text{Nm}^2$
Permeability of Free Space	μ_0	$4\pi \times 10^{-7} \text{ Tm/A}$
Coulomb constant	k	$8.99 \times 10^9 \text{ Nm}^2/\text{C}^2$
Speed of light in a vacuum	c	$3.00 \times 10^8 \text{ m/s}$
Fundamental charge	e	$1.602 \times 10^{-19} \text{ C}$
Planck's constant	h	$6.626 \times 10^{-34} \text{ Js}$
Electron mass	m_e	$9.11 \times 10^{-31} \text{ kg}$
Proton mass	m_p	$1.67265 \times 10^{-27} \text{ kg}$ 1.007276 amu
Neutron mass	m_n	$1.67495 \times 10^{-27} \text{ kg}$ 1.008665 amu
Atomic Mass Unit	amu	$1.66 \times 10^{-27} \text{ kg}$ $931.5 \text{ MeV}/c^2$
Gravitational constant	G	$6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$
Stefan-Boltzmann constant	σ	$5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$
Universal gas constant	R	$8.314 \text{ J/mol} \cdot \text{K}$ $0.082057 \text{ L} \cdot \text{atm/mol} \cdot \text{K}$
Boltzmann's constant	k_B	$1.38 \times 10^{-23} \text{ J/K}$
Speed of Sound (at 20°C)	v	343 m/s
Avogadro's number	N_A	$6.022 \times 10^{23} \text{ atoms/mol}$
Electron Volts	eV	$1.602 \times 10^{-19} \text{ J/eV}$
Distance Conversion	miles $\rightarrow$ meters feet $\rightarrow$ meters	1.00 mile = 1609 meters 1.00 foot = 0.3048 meters
Rydberg Constant	R_∞	$1.097 \times 10^7 \text{ m}^{-1}$
Standard Atmospheric Pressure	1 atm	$1.013 \times 10^5 \text{ Pa}$
Density of Pure Water	ρ_{water}	1000.0 kg/m^3